

Lecture 16: Ray theory

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Outline and Motivation

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- We will develop **ray theory**, an approximation for when the material changes slowly compared to the wavelength.
- The key idea is that waves behave **locally** like plane waves, but their wavefronts curve and amplitudes change as they propagate.
- This is a "high-frequency" approximation, analogous to **geometric optics** in electromagnetism.

Wave Propagation Regimes

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Averaging

$L/\lambda \ll 1$ The wave doesn't resolve the fine details and "sees" an averaged medium.

Ray Theory Ansatz (1/2): Intuition

We assume the solution in the heterogeneous region takes a form similar to a plane wave, but with position-dependent terms.

$$u_i(\mathbf{x}, t) = f[t - T(\mathbf{x})]a_i(\mathbf{x})$$

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Travel Time $T(\mathbf{x})$

The phase is now a non-linear function of position. The surfaces $T(\mathbf{x}) = \text{constant}$ are the **wavefronts**. The gradient of the travel time defines the local **slowness vector**:

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Amplitude $a_i(\mathbf{x})$

The amplitude and polarisation vector can now vary in space, accounting for focusing/defocusing and changes in particle motion.

Ray Theory Ansatz (2/2): The Ray Series

To be more rigorous, we seek a solution as a series expansion in inverse powers of frequency, ω . In the frequency domain, this is:

$$u_i(\mathbf{x}, \omega) = \left(\sum_{n=0}^{\infty} \frac{a_i^n(\mathbf{x})}{(i\omega)^n} \right) e^{-i\omega T(\mathbf{x})}$$

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- This is an **asymptotic expansion**, which is accurate for high frequencies ($\lambda \ll L$) but may not converge in the usual sense.

Hierarchy of Equations

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Highest Order $((i\omega)^2)$: The Eikonal Equation

This equation determines the travel time, $T(\mathbf{x})$.

$$[A_{ijkl}p_j p_l - \rho\delta_{ik}]a_k^0 = 0$$

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We will focus on solving the first and most important of these: the Eikonal equation.

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The Eikonal Equation

$$\det[\mathbf{\Gamma}(\mathbf{x}, \mathbf{p}(\mathbf{x})) - \mathbf{I}] = 0$$

This is a first-order, non-linear partial differential equation for the travel time $T(\mathbf{x})$, since $\mathbf{p} = \nabla T$.

Solving the Eikonal Equation

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- Let $c_k(\mathbf{x}, \hat{\mathbf{p}})$ be the phase speed for the k -th wave type in direction $\hat{\mathbf{p}}$ at point $\mathbf{x}$. The equation becomes:

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To solve this PDE, we use a powerful technique called the **method of characteristics**.

Method of Characteristics (1/2): Rays

The method of characteristics turns the PDE for $T(\mathbf{x})$ into a system of ODEs that trace out rays.

- First, we define the **Ray Hamiltonian** for the k -th wave type:

$$H_k(\mathbf{x}, \mathbf{p}) = \frac{1}{2} \|\mathbf{p}\|^2 c_k[\mathbf{x}, \hat{\mathbf{p}}]^2$$

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The eikonal equation is then simply $H_k = 1/2$.

- The path of a ray, $\mathbf{x}(\sigma)$, and its evolving slowness vector, $\mathbf{p}(\sigma)$, are found by integrating **Hamilton's equations**:

Hamilton's Ray Tracing Equations

$$\frac{dx_i}{d\sigma} = \frac{\partial H_k}{\partial p_i} \quad \text{and} \quad \frac{dp_i}{d\sigma} = -\frac{\partial H_k}{\partial x_i}$$

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Physical Meaning

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Caustics

- A **caustic** is a location where rays cross.
- At a caustic, the travel time function is multi-valued, and the ray theory amplitude a_k^0 becomes infinite.
- This is a breakdown of the approximation. In reality, amplitudes at caustics are very large but finite.

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- The highest-order equation is the **eikonal equation**, a PDE for the travel time $T(\mathbf{x})$.
- The eikonal equation is solved by the **method of characteristics**, which involves integrating **Hamilton's equations** to trace **rays**.
- The parameter along a ray is the travel time. A **caustic** is formed where rays cross, and is a point where simple ray theory breaks down.